Spring Boot Actuators

Actuators are production-ready monitoring and management tools that can be easily added to a Spring Boot application. They provide endpoints to monitor, manage, and gather metrics from your app.

```xml
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-actuator</artifactId>
</dependency>
```

GET Operations

- Visit `http://localhost:8080/actuator` to see available endpoints:

```json
{
  "_links": {
    "self": {
      "href": "http://localhost:8080/actuator",
      "templated": false
    },
    "health": {
      "href": "http://localhost:8080/actuator/health",
      "templated": false
    },
    "health-path": {
      "href": "http://localhost:8080/actuator/health/{*path}",
      "templated": true
    }
  }
}
```

- Control which endpoints are exposed via `application.yml`:

```yaml
management:
  endpoints:
    web:
      exposure:
        include: loggers,info,health
```

- To **expose all endpoints**:

```yaml
management:
  endpoints:
    web:
      exposure:
        include: "*"
```

/health Endpoint

- Provides an **overview of application health**
- Returns statuses: `UP`, `DOWN`, or `UNKNOWN`
- Can include **detailed health indicators** (database, disk space, external services)
- Supports **custom health checks** via `HealthIndicator` beans

View Health Endpoint →

/metrics Endpoint

- Publishes **OS, JVM, and application-level metrics**
- Example data: memory, heap, processors, threads, classes loaded/unloaded, thread pools, HTTP metrics

View Metrics Endpoint →

/beans Endpoint

- Provides **detailed information about all beans** in the Spring context

View Beans Endpoint →

/loggers Endpoint

- Shows **current and configured logging levels**
- Allows **runtime updates** to logging configuration

View Loggers Endpoint →

POST Operations

- Enable POST for runtime environment changes:

```yaml
management:
  endpoint:
    env:
      post:
        enabled: true
```

- Example: Update ROOT logging level to TRACE at runtime:

```bash
curl -X POST "http://localhost:8080/actuator/loggers/ROOT" \
  -H "Content-Type: application/json" \
  -d '{
        "configuredLevel": "TRACE"
      }'
```

- Allows **dynamic debugging and tuning** without restarting the app

Key Takeaways

- Actuators bring **production-ready monitoring** with minimal effort
- Endpoints provide **health, metrics, beans, loggers**, and more
- Supports **GET for monitoring** and **POST for runtime configuration**

Resources

- Spring Boot Actuator Guide
- Custom Info in Info Endpoint